

Response of STF Influenced Lattice Structures Under Complex Loading

Background

Protective gear is vital for humans, as it needs to meet specific requirements to withstand wear and tear. Currently, these systems rely on metals and fibers that add uncomfortable weight, along with being expensive (Alexe, 2024). **Shear thickening fluids** (STF) have been an emerging area for these types of problems. Due to the adaptable quality STFs give, they are suitable for protective applications, as protection foams and fabrics that are engulfed in STFs are more resistant to unexpected attacks than the untreated objects (Liu, 2023). These tests are crucial before fully emerging in appliances, as multiaxial loading simulates the real world.

Rationale

- ❖ This research is important because shear thickening fluids stiffen under high shear rates to absorb sudden impact energy, making them highly suitable for protective gear (Alexe, 2024), yet there is a significant gap in the literature regarding their integration with metamaterials—particularly under dynamic, multi-directional loading conditions beyond the uniaxial focus of existing studies (Wei, 2022)—and no established experimental research currently involves STF behavior with lattice geometry during impact events.

Research/Eng Goal

How can the integration of shear thickening fluids and metamaterial structures enhance energy absorption and mechanical performance under complex loading?

Null Hypothesis (H0)

- The integration STF into metamaterial lattice structures does not result in a statistically significant improvement in energy absorption or mechanical performance under complex loading conditions compared to metamaterial structures without.

Experimental Hypothesis (H1)

- The integration STF with metamaterial lattice structures significantly enhances energy absorption and mechanical performance under complex loading conditions compared to metamaterial structures without STFs.

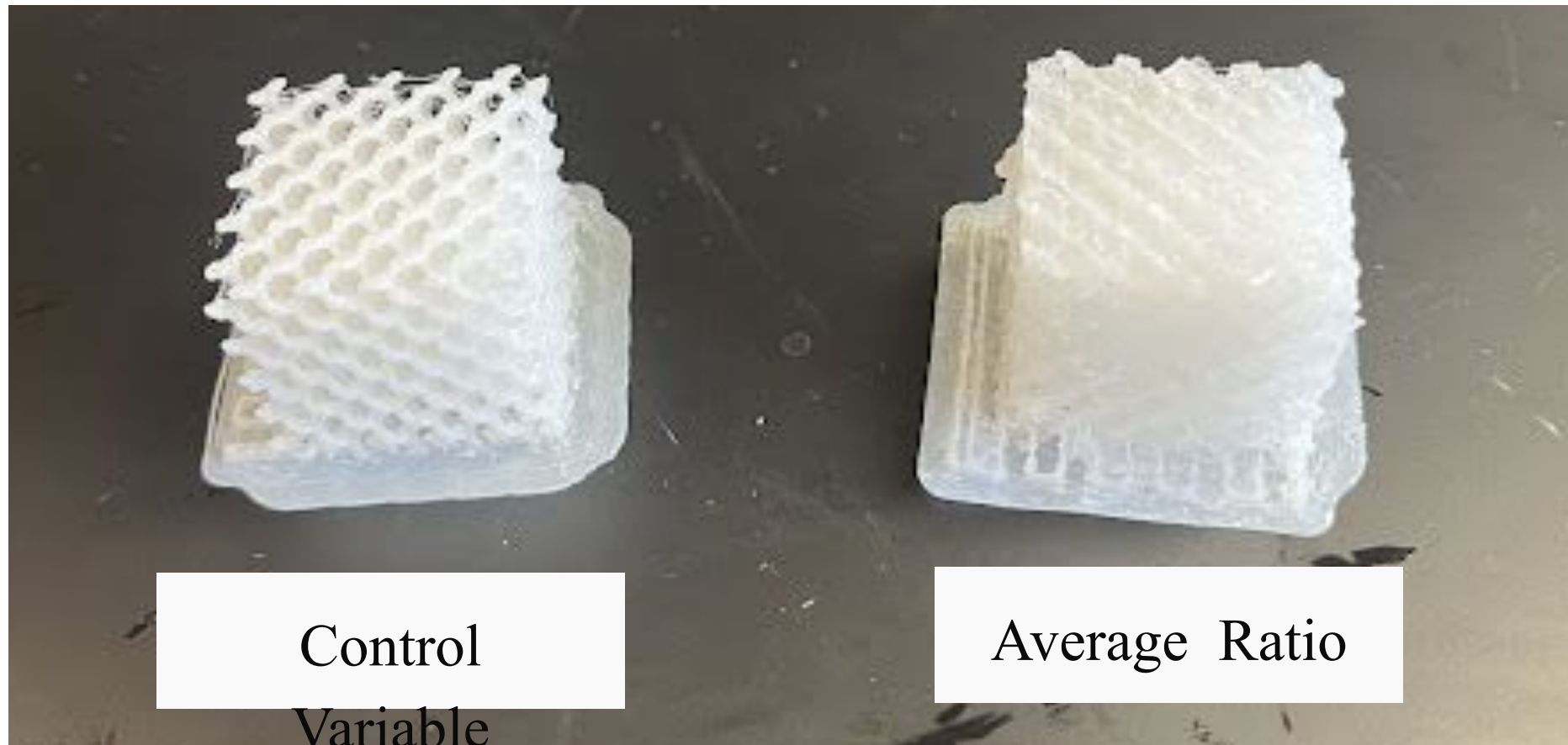
Methods

Figure 1
STF Mixture

STF Type	Silica (g)	PEG 400 (ml)	Ratio	Evaluation
Low	0.126g	1ml	10 wt.%	Fluid
Average	0.484g	1ml	30 wt.%	Average viscosity
High	1.13g	1ml	50 wt.%	Dense viscosity
Control	0g	0ml	0 wt.%	Newtonian fluid

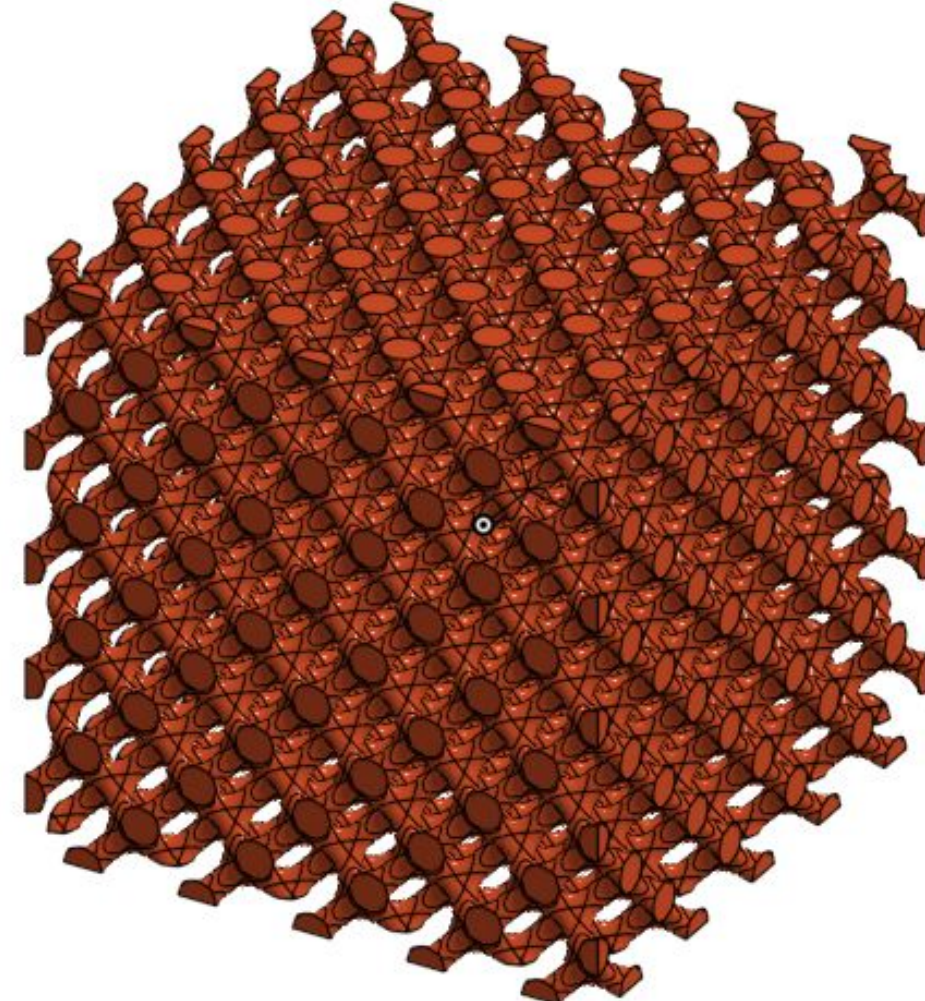
Note. Ratio of silica in grams to PEG in ml (Serra, 2024)

Figure 3
STF Treated Lattice Structure vs. Untreated Lattice Structure



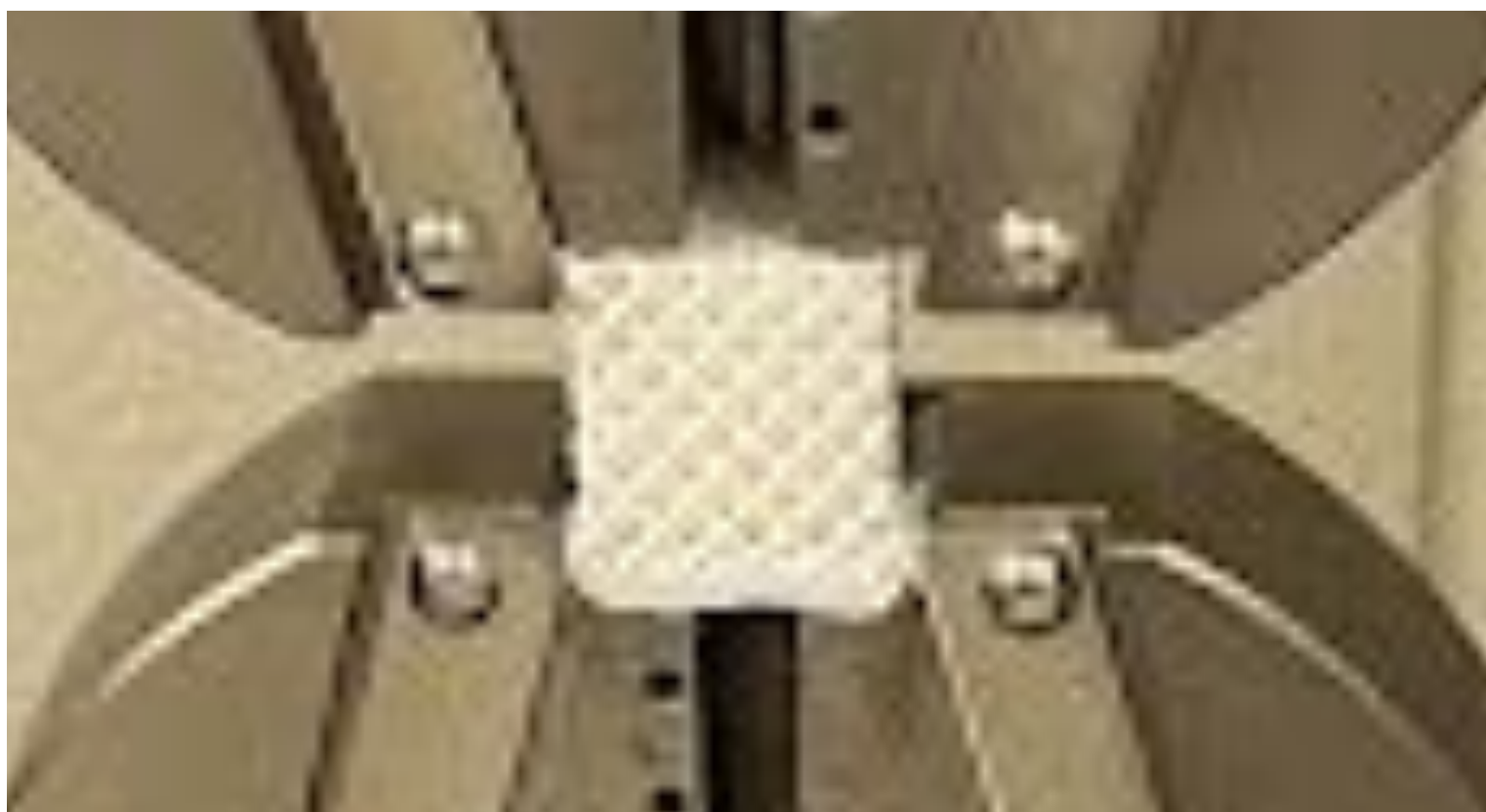
Note. The left model is the control without STF, and the right model contains STF.

Figure 2
Digital Lattice Structure Model



Note. 3D model programmed on OnshapeCAD

Figure 4
UTM Testing



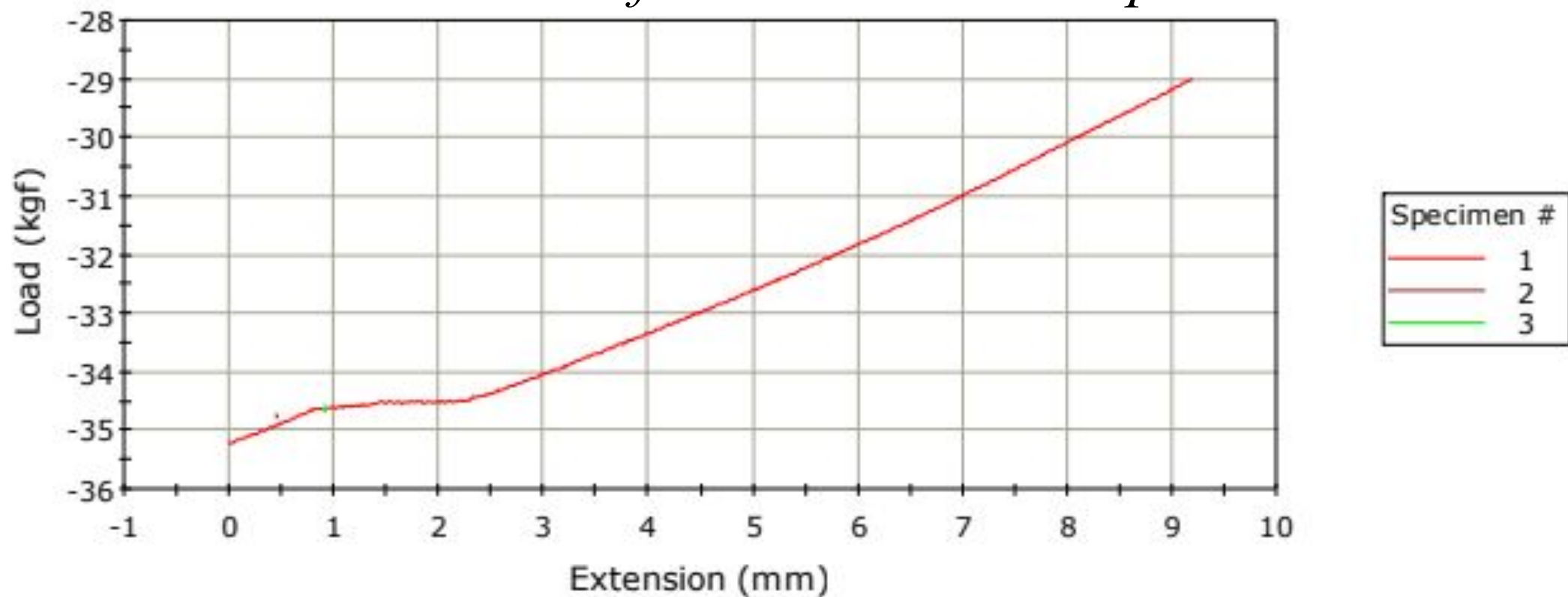
Note. Lattice specimen mounted in the universal testing machine

Conclusion

- The infused shear-thickening fluid (STF) increases the lattice's resistance to deformation which improves the overall mechanical stability.
- STF eliminates major buckling behavior typically observed in unfilled lattice structures.
- The compressive response becomes smooth and nearly linear after initial seating, the load–deformation curve rises steadily (from approximately −34.7 kgf to −28.5 kgf) which suggests that these structures enhanced structural response.
- The absence of a load drop at the end suggests a significantly increased load capacity.
- Overall, combining STF with metamaterial lattice structures enhances energy absorption and mechanical performance under realistic loading conditions.
- The study helps close a key research gap and supports the development of advanced protective systems.

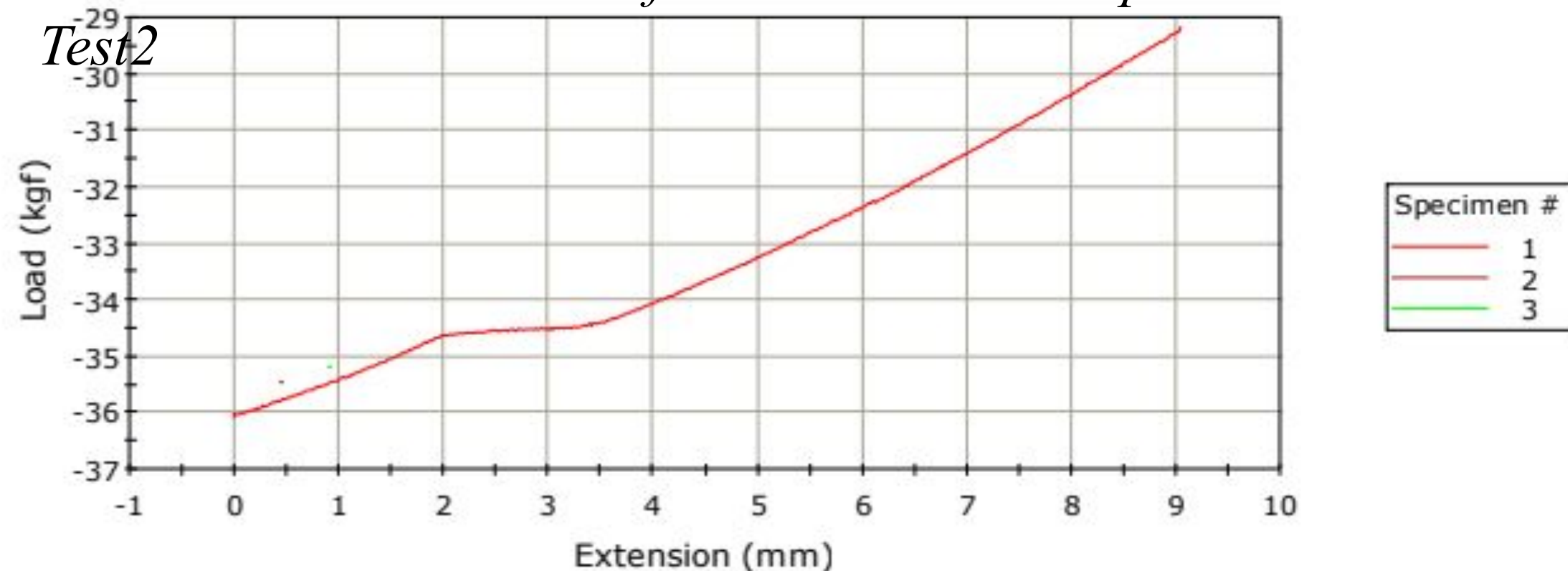
Results

Figure 5
Load vs. Extension Curve for 30 wt.% Lattice Specimen



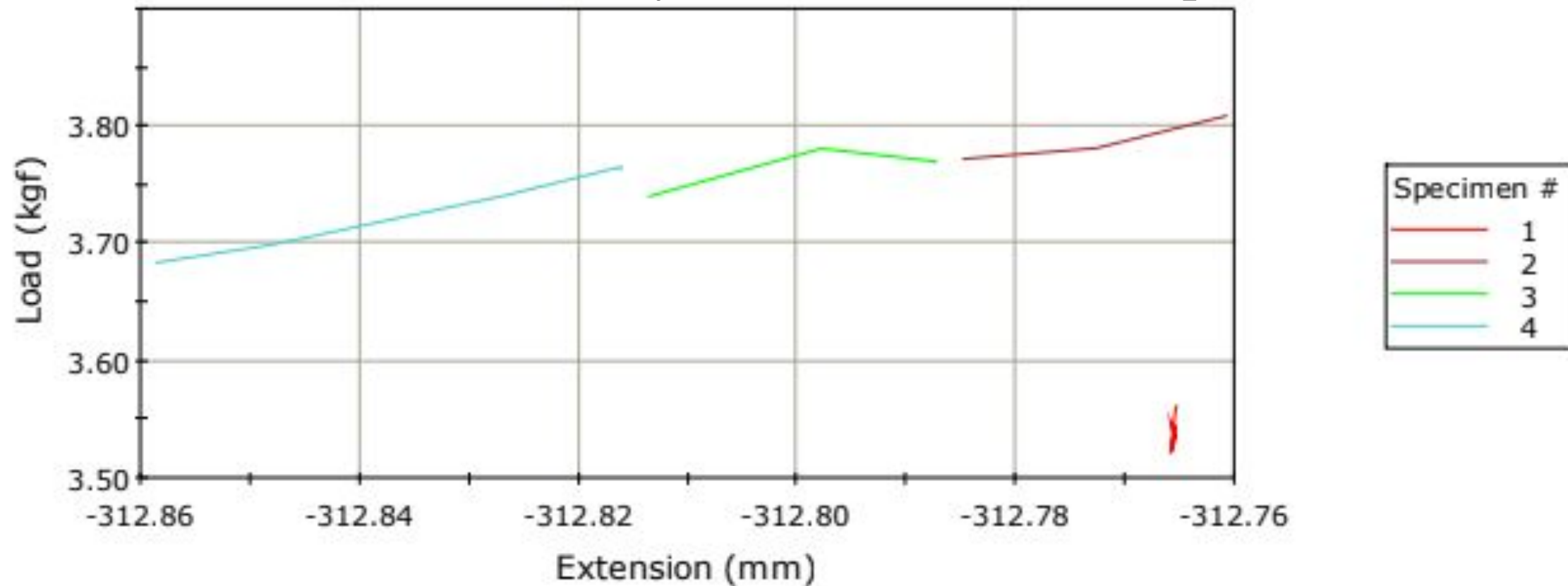
Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 6
Load vs. Extension Curve for 30 wt.% Lattice Specimens



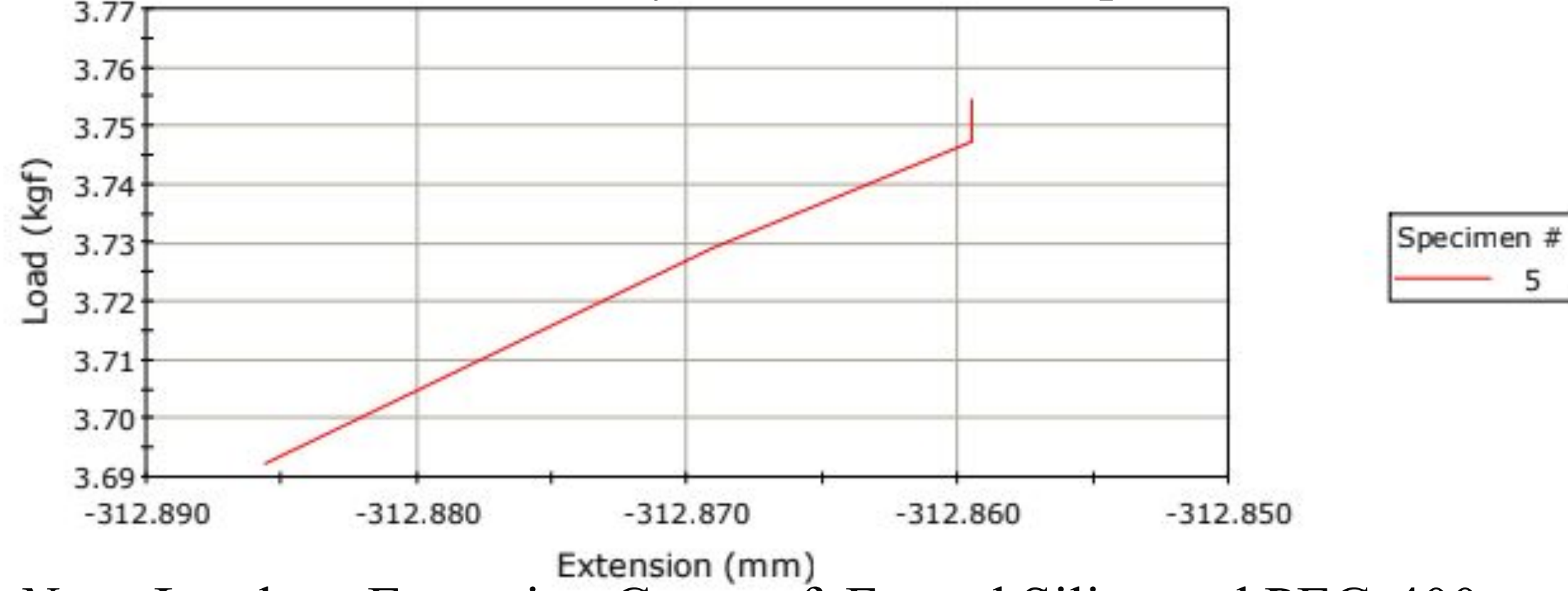
Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 7
Load vs. Extension Curve for ALL wt.% Lattice Specimens



Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 8
Load vs. Extension Curve for 0 wt.% Lattice Specimens



Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Future work

- Requires additional STF mixtures with a wider range of silica-to-PEG ratios to further explore how composition influences mechanical performance.
- Additional data using the Universal Testing Machine under multiple loading conditions, including standard compression, ballistic loading, and creep testing will provide more comprehensive and quantitative data on the effectiveness of STF infusion.

References

- Alexe, F., Sau, C., Iorga, O., Toader, G., Diacon, A., Rusen, E., Lazaroaie, C., Gingham, R. E., Tiganescu, T. V., Teodorescu, M., & Sobetkii, A. (2024). Experimental Investigations on Shear Thickening Fluids as “Liquid Body Armors”: Non- Conventional Formulations for Ballistic Protection. *Polymers*, 16(16), 2305. <https://doi.org/10.3390/polym16162305>
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